

Crystallization behavior of glasses in the system of Na_2O – CaO – MgO – Fe_2O_3 – Al_2O_3 – SiO_2 with high contents of nickel oxide

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Abstract

The structural behavior of nickel oxide in glassy and glass-ceramic materials, obtained in the system of Na_2O – CaO – MgO – Fe_2O_3 – Al_2O_3 – SiO_2 , was investigated. The influence of the NiO content on the vitrification, crystallization, structure and exploitation properties of two model compositions, with different ratios $[\text{CaO}]/[\text{MgO}]$ was analyzed. On the basis of DSC and XRD data, it is shown that NiO promoted the formation of bunsenite crystals, as nuclei for crystallization. On the other hand, NiO promoted formation of pyroxenes even for compositions with low MgO contents, which formed gehlenite without NiO admixtures. It is shown that in the composition with relatively high MgO contents, NiO could participate in the formation of two types of pyroxenes with the structure and chemical composition similar to $(\text{MgO}_{0.4}\text{NiO}_{0.6})(\text{CaO}_{0.9}\text{NiO}_{0.1})\text{Si}_2\text{O}_6$ and diopside–hedenbergite solid solutions. The optimal contents of NiO in both model compositions was about of 7 wt%, since higher contents reduced the exploitation properties. The glass-ceramics with optimal contents of NiO were also produced using Ni bearing galvanic slurry and coal ash; the resulting materials showed similar exploitation properties to those mentioned above.

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1. Introduction

The amounts of industrial and municipal wastes bearing ecologically harmful elements are growing rapidly; moreover, according to national and international regulations this entails high costs for its disposal, economically and ecologically. One potential use of such harmful wastes may be as raw materials for the production of useful and at the same time environmentally friendly materials, which will be highly effective, mechanically strong, well-formed and cheaper than commercially available materials. Some

of such wastes that may serve these purposes are the slurries spent from galvanic processes.

The slurries obtained from the neutralization of spent galvanic solutions used to form Zn and Cu coatings are considered as very harmful wastes due to the presence of cyanides and cannot be applied in manufacturing of glassy and glass-ceramic materials. However, the slurries used to obtain Ni and Cr coatings, which also are harmful but do not contain cyanides, are potentially interesting to be utilized in glass-ceramics.

On the other hand, pyroxene glass-ceramics is a promising type of material, which could be developed from different industrial wastes [1–10]. Among the properties that make such glass-ceramic materials attractive are their high mechanical strength and excellent chemical durability [6,7]

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and the low sensitivity of their structure and properties to variations of the chemical compositions of the wastes used [8–10].

The pyroxene glass-ceramic materials produced from batches containing galvanic slurries seem as quite promising ways to use other industrial wastes like ash or slag (sources of SiO_2 , Al_2O_3 , Fe_2O_3 , CaO , MgO). These material, thus, could have a highly positive impact in the management and disposal stages of other industrial wastes.

NiO (Ni_2O_3) usually is considered a nucleation agent for regulated crystallization of silicate glasses, it is commonly introduced as an admixture in relatively small percentages (0.5–2 wt%) [11,12]. Moreover, the crystallization behavior of glasses with stoichiometric compositions of $2\text{NiO} \cdot 2\text{Al}_2\text{O}_3 \cdot 5\text{SiO}_2$ and $\text{NiO} \cdot \text{MgO} \cdot 2\text{Al}_2\text{O}_3 \cdot 5\text{SiO}_2$ were investigated to produce glass-ceramic materials with NiAl_2O_4 crystalline phase [13]. However, there is no published research data on the behavior of NiO in glasses of pyroxene compositions with high contents of this oxide.

Under the above considerations, this research undertakes the synthesis and characterization of the pyroxene glass-ceramics obtained in the model system $\text{Na}_2\text{O}-\text{CaO}-\text{MgO}-\text{Fe}_2\text{O}_3-\text{Al}_2\text{O}_3-\text{SiO}_2$ with different NiO contents. Furthermore, the obtained results are used to investigate the crystallization behavior of high NiO containing glasses and the properties of glass-ceramic materials produced with industrial wastes such as galvanic slurry and coal ash from a power plant.

2. Experimental

The chemical composition of all materials used is presented in Table 1. Two basic glass compositions were investigated (Table 2). The prepared compositions were selected from typical glass-ceramics materials produced in the system of $\text{R}_2\text{O}-\text{CaO}-\text{MgO}-\text{Fe}_2\text{O}_3-\text{Al}_2\text{O}_3-\text{SiO}_2$ in accordance with previous research data [1–12,14,15]. The compositions were characterized by fairly similar contents of SiO_2 , Al_2O_3 , Fe_2O_3 and Na_2O ; they had different $[\text{CaO}]/[\text{MgO}]$ ratios, of 2.7 (compositions 1.1.x) and 23.7 (compositions 1.2.x), influencing the rate of crystallization and phase composition of the resulting glass-ceramics [14,15].

For Series 1, batches of the two model compositions (1.1.0 and 1.2.0) were prepared using traditional raw mate-

rials as sand, dolomite, limestone; Fe_2O_3 , Al_2O_3 and Na_2CO_3 (Aldrich, chemical grade of 99+%). The influence of NiO , on the devitrification behavior of the glasses and the structure and properties of the glass-ceramics obtained, was investigated on Series 1 by means of admixtures of 4, 7 and 10 wt% of NiO (in excess of the 100% of the batch compositions 1.1.0 and 1.2.0). $\text{Ni}(\text{NO}_3)_2$ (Aldrich, chemical grade of 99+%) was used for such purposes.

Glasses from Series 1 (number 1.1.7 and 1.2.7) were selected as more promising and were further produced and investigated in Series 2. For the latter, the NiO source was replaced for a spent dried galvanic slurry (company Cromocentro S.A. de C.V., Saltillo, Mexico), it was neutralized to $\text{pH} = 7$, by adding $\text{Ca}(\text{OH})_2$ aqueous suspensions, filtrated and dried at 98°C for 4 h. Taking into account that the galvanic slurries contain other oxides present in the glass compositions previously selected, the batch of the compositions for Series 2 were recalculated. The composition of raw materials for Series 2 is indicated in Table 3. Composition 2.1.7 was prepared with galvanic slurry and the traditional raw materials indicated above. Composition 2.2.7 was a combination of two types of waste, namely galvanic slurry and coal ash, with small additions of Na_2CO_3 .

The glasses were fused at 1450°C for 2 h and formed by quenching into graphite moulds. The temperature of crystallization of the produced glasses was selected on the basis of the DSC (BAHR STA503) data, after heating at $10^\circ\text{C}/\text{min}$. The structure and chemical composition of the glass-ceramics obtained after the thermal treatments of the glassy specimens at different temperatures were investigated by electron scanning microscopy (Philips XL30E-SEM, equipped with an X-ray micro-analyzer EDAX Pegasus). The samples were cut with a diamond disc, polished with diamond paste and carbon coated to make the surfaces conductive. The phase composition of glass-ceramic powder (grain size of 0.05–0.15 mm) was analyzed by X-ray diffractometry (Philips X'Pert-MPD).

The flexural mechanical strength was characterized by testing 10 samples (size $100 \times 20 \times 7$ mm) of each glass-ceramic material by means of the three-point method (Controls-15 KN, model 65-L1301). The chemical resistance of the glass and glass-ceramics were measured in accordance with Russian standard GOST 10134-82. Grains of material

Table 1
Chemical composition of the used raw materials (EDS analysis of areas)

Raw material	Contents of oxide (wt%)								
	SiO_2	Al_2O_3	Fe_2O_3	CaO	MgO	Na_2O	NiO	Cr_2O_3	LOI
Sand	99.5	0.2	–	0.1	–	–	–	–	0.3
Dolomite	2.0	0.8	0.1	33.8	20.5	–	–	–	43.6
Limestone	1.5	0.2	0.2	56.0	–	–	–	–	42.8
Fe_2O_3	–	–	99.5	–	–	–	–	–	0.5
Al_2O_3	–	99.5	–	–	–	–	–	–	0.5
Na_2CO_3	–	–	–	–	–	58.5	–	–	41.5
Galvanic slurry	–	–	14	24	–	–	38.1	1.1	22.8
Coal ash	50.1	13.3	8.4	26	2.1	–	–	–	0.2

Table 2
Chemical composition of the glasses investigated (EDS analysis of areas)

Series	Number	Admixture of NiO in excess of 100 wt%	Contents of oxide (wt%)							
			SiO ₂	Al ₂ O ₃	Fe ₂ O ₃	CaO	MgO	Na ₂ O	NiO	Cr ₂ O ₃
1	1.1.0	0	41.5	11.7	10.3	22.4	8.9	5.2	–	–
	1.1.4	4	41.4	11.2	9.9	21.8	8.3	3.5	3.9	–
	1.1.7	7	41.1	11.0	8.5	21.4	7.9	4.0	6.1	–
	1.1.10	10	40.4	10.6	7.4	21.5	7.6	3.7	8.8	–
	1.2.0	0	40.7	12.3	10.5	30.6	1.3	4.6	–	–
	1.2.4	4	40.3	12.1	9.5	28.1	1.5	4.4	4.1	–
	1.2.7	7	40.2	11.1	8.6	28.8	1.6	3.4	6.3	–
	1.2.10	10	39.2	11.2	8.2	27.8	1.2	3.1	9.3	–
2	2.1.7 (analog to 1.1.7)	With galvanic slurry	41.0	11.0	8.5	21.5	7.8	3.9	6.2	0.2
	2.2.7 (analog to 1.2.7)	With galvanic slurry	41.4	11.0	8.6	26.5	1.8	3.4	7.2	0.2

Table 3
Batch compositions used in the Series 2

Number of batch	Contents of raw materials in the batch (wt%)						
	Sand	Al ₂ O ₃	Fe ₂ O ₃	Dolomite dust	Dried galvanic slurry	Coal ash	Na ₂ CO ₃
2.1.7	30.9	8.6	4.5	37.2	11.7	–	7.1
2.2.7	–	–	–	–	17.3	78.2	4.5

(three samples) with particle size of 0.5–0.8 mm (10 g) were treated at 98 °C in 100 ml of: distilled water (water resistance) for 5 h, 1 N HCl aqueous solution (acid resistance) for 3 h or 1 N NaOH aqueous solution (alkali resistance) for 3 h. The results were expressed as weight losses in percent. After the water resistance tests, aqueous solutions were obtained and analyzed with an emission spectrophotometer (Perkin–Elmer 2380) to estimate the contents of Ni and Cr in the leachates and compare them with the limits established in the land disposal restrictions (LDR) of the US Environmental Protection Agency (FR 62079, May 26, 1998). The density of obtained materials was measured by pycnometric method.

3. Results

The measured chemical composition of the model glasses with different admixtures of NiO and glasses produced with industrial wastes are presented in Table 1, the DSC data obtained for such glasses are reported in Fig. 1. The glasses with different [CaO]/[MgO] ratios showed similar temperatures of glass transition (about of 680–690 °C). However, for the glass No 1.1.0 (higher MgO content in comparison with the glass No 1.2.0), the exothermal peak of crystallization appeared at lower temperature (990 and 1017 °C, respectively).

The admixture of [NiO], in the compositions of Series 1.1.x (Fig. 1(a)), initially resulted in a reduction of the temperature of the principal exothermal peak: for composition 1.1.0 it was at 990 °C, and shifted to 856 °C for composition 1.1.4. In the composition 1.1.7 this peak shifted to 860 °C and an additional peak, weakly apparent in the

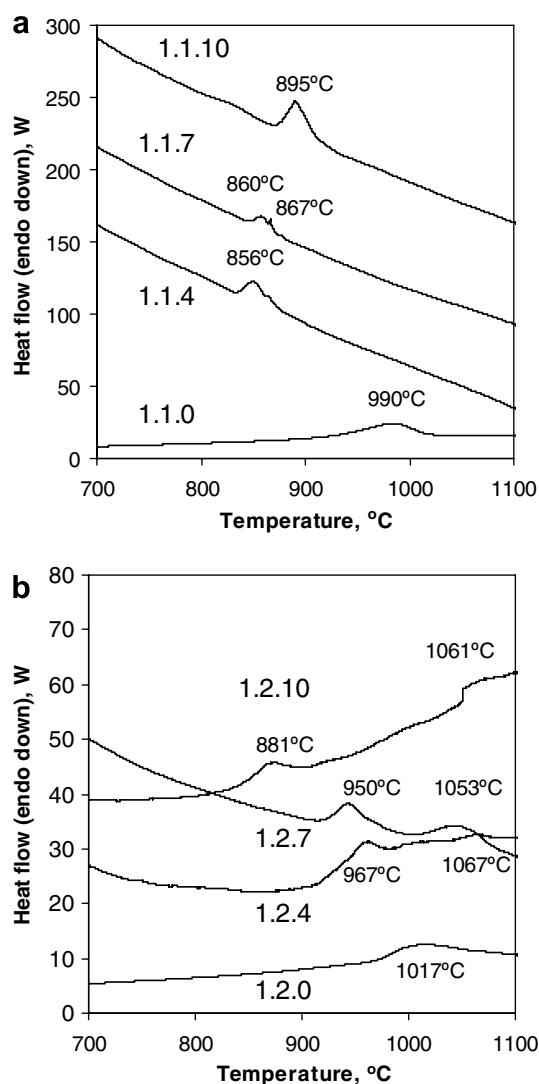


Fig. 1. DSC data for the glasses based on the composition of Series 1.1.x (a) and 1.2.x (b).

composition 1.1.4, could be identified at 867 °C. Further increase in [NiO] showed only one exothermal peak at 895 °C (composition 1.1.10).

In Series 1.2.x, for glass No 1.2.0 the exothermal peak appeared in the range of 980–1050 °C (maximum at 1017 °C); and the introduction of gradually increasing [NiO] contents resulted in shifting such peak down to 967 °C (composition 2.1.4), 950 °C (composition 1.2.7) and 881 °C (composition 1.2.10).

The above data indicates that in the NiO free glass compositions the crystalline phases can only appear at about 1000 °C (compositions 1.1.0 and 1.2.0); it also suggests that for the NiO containing glass compositions the thermal treatment at 950 °C could support the crystallization process. Heating at 10 °C/min and soaking for 2 h were

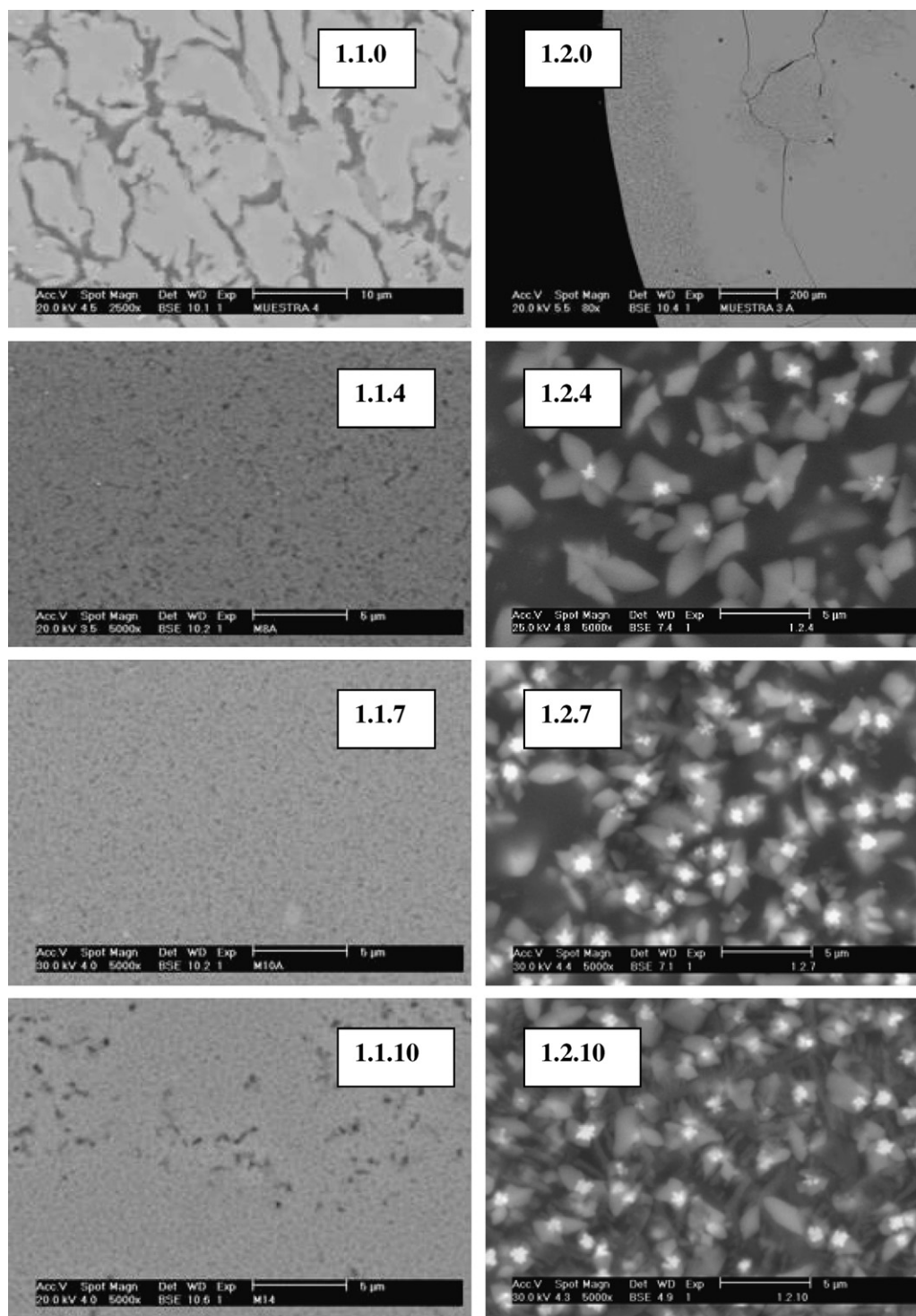


Fig. 2. Micrographs of glass-ceramics obtained in Series 1.1.x and 1.2.x by the thermal treatment at 1000 °C (No 1.1.0 and 1.2.0) and 900 °C (magnification 5000x).

selected as typical conditions of regulated crystallization for pyroxene glass-ceramics [1–14]. After the crystallization the temperature was decreased to 680 °C at 10 °C/min, the specimens were soaked for 0.5 h for annealing; further cooling down to room temperature was performed at 30°/min. The temperature of annealing was selected taking into account the temperature of glass transition identified at 650 ± 20 °C for all glasses investigated.

The electron micrographs of the materials, obtained after such thermal treatment, are reported in Fig. 2. The basic NiO free glass compositions formed glass-ceramics at 1000 °C; however the crystalline phase (average size of crystals about of 10 μm) was much better developed in the bulk of the material with higher content of MgO (composition 1.1.0), and only some groups of crystals were noted in the bulk of the material of composition 1.2.0. The introduction of NiO in both basic compositions promoted the formation of glass-ceramic materials with bulk crystallization at 950 °C.

For the glass compositions with 4 wt% NiO, those with higher content of MgO (composition 1.1.4) promoted an intensive crystallization with a fine-crystalline structure of glass-ceramics (average size of crystals about of 0.1 μm); whereas the glass 1.2.4 (lower MgO content) formed a glass-ceramics with crystals sized in the range of 3–5 μm and characterized by higher content of a vitreous phase. Increased NiO contents to 7 and 10 wt% (compositions 1.1.7 and 1.2.7) promoted a more intensive crystallization of both glass compositions.

The EDS spot analysis of different crystals in the glass-ceramics 1.1.4, 1.1.7 and 1.1.10 indicated NiO contents (wt%) that varied in the ranges of: 9.2–19.6, 5.6–28.9 and 11.2–21.1, respectively; whereas the crystals of glass-ceramics 1.2.4, 1.2.7 and 1.2.10, showed more stable NiO contents in the range of 5.4–9.8 wt%.

The XRD data, reported in Figs. 3 and 4, allowed the identification of the crystalline phases formed in the produced materials. The basic compositions (1.1.0) crystallized forming mainly a pyroxene-like phase similar to diopside–hedenbergite solid solutions.

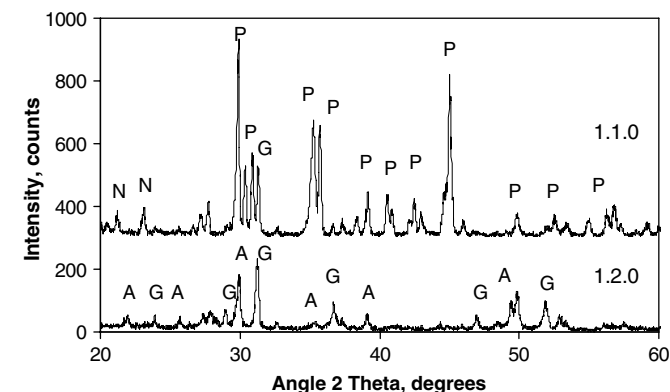


Fig. 3. X-ray diffractograms of the glass-ceramics (basic compositions 1.1.0 and 1.2.0, 1000 °C, 2 h). P – pyroxene-like crystalline phase similar to diopside–hedenbergite solid solutions, G – gehlenite–akermanite solid solutions, A – augite, N – nepheline.

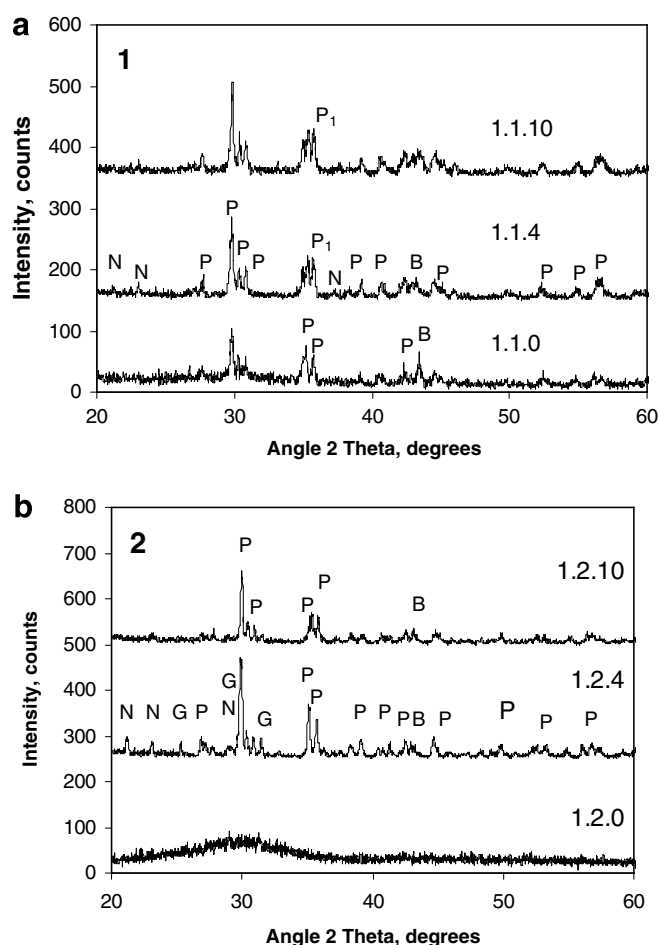


Fig. 4. X-ray diffractograms of glass-ceramic materials produced from glasses of Series 1 treated at 950 °C for 2 h. P – pyroxene-like crystalline phase similar to diopside–hedenbergite solid solutions, P₁ – characteristic reflection of the pyroxene-like crystalline phase similar to $(\text{Mg}_{0.4}\text{Ni}_{0.6})(\text{Ca}_{0.9}\text{Ni}_{0.1})\text{Si}_2\text{O}_6$, N – nepheline, G – gehlenite–akermanite solid solutions, B – bunsenite (NiO).

hedenbergite solid solutions, as well as gehlenite–akermanite solid solutions and nepheline (Fig. 3). The composition 1.2.0, similarly treated, showed the presence of gehlenite (ICDD 79-2422) and pyroxene-like augite (ICDD 79-3753), as the main crystalline phases.

XRD data shown in Fig. 4 indicates that for the glass-ceramics from Series 1.1.x (high [MgO]), increased NiO contents supported the formation of a pyroxene-like crystalline phase with a structure similar to $(\text{Mg}_{0.4}\text{Ni}_{0.6})(\text{Ca}_{0.9}\text{Ni}_{0.1})\text{Si}_2\text{O}_6$ and characterized with a set of reflections in the 2θ range of 35°–36° (35.31°, 35.72° and 35.80°, ICDD 83-0526) different to those typical for the pyroxene-like diopside–hedenbergite solid solutions (35.27° and 35.71°, ICDD 83-2016, 83-0088). On the other hand, for Series 1.2.x (low [MgO]), the introduction of NiO supported intensive crystallization of the pyroxene-like diopside–hedenbergite solid solutions instead of gehlenite and augite; the former was the only crystalline phase for composition 1.2.10, while composition 1.2.4 also indicated the presence of some gehlenite and nepheline. Additionally,

bunsenite (NiO) was found in the glass-ceramics obtained from both NiO containing Series 1.1.x and 1.2.x; the bunsenite crystals from compositions 1.2.x were sufficiently large to identify them by EDS spot analysis (Fig. 2).

The influence of NiO contents on the properties of the obtained glass-ceramic materials is reported in Figs. 5 and 6. The introduction of NiO in Series 1.1.x (high [MgO]) increased the chemical resistance of the glass-ceramics in water and alkaline aqueous solution (Fig. 5). For compositions 1.2.x (low [MgO]), the water resistance also increased but the resistance in both alkali and acid aqueous solutions decreased. For all NiO additions, the glass-ceramic materials from Series 1.1.x showed higher alkali resistance and lower water resistance in comparison with those from Series 1.2.x; the resistance in acid aqueous solution was similar for both Series.

In Series 1.1.x, the additions of up to 7 wt% NiO did not influence the density of the resulting glass-ceramics (Fig. 6); however higher NiO contents increased the density. For the

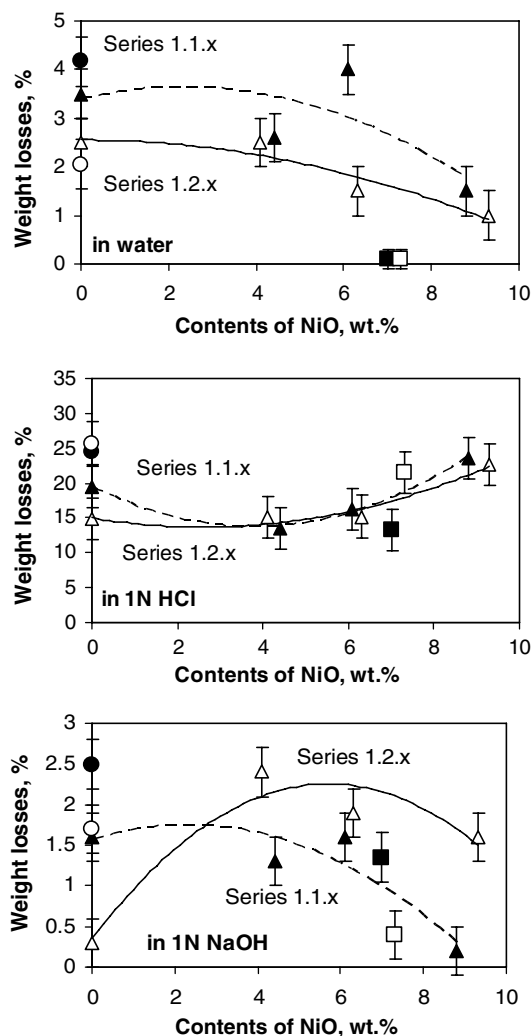


Fig. 5. Chemical resistance of glass-ceramic materials obtained with different admixtures of NiO (950 °C, 2 h). Data for the glasses and glass-ceramics from Series 1.1.x and 1.2.x are marked as ●, ■ and ○, □ respectively. The lines are presented as guides for the eye.

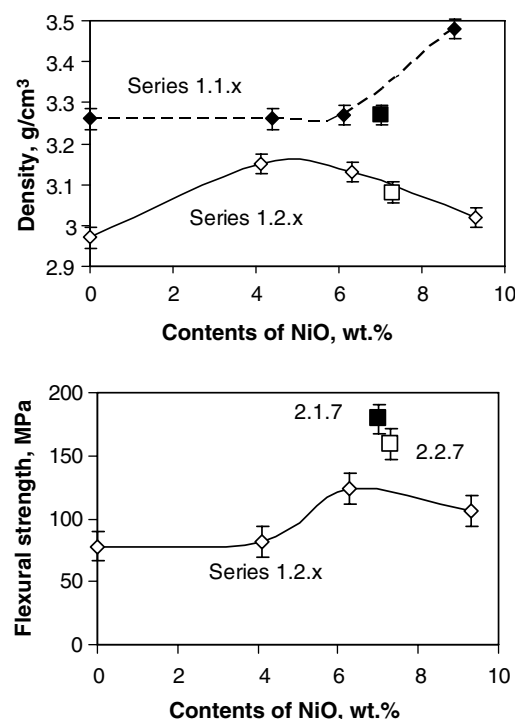


Fig. 6. Density and flexural strength of glass-ceramics produced with different admixtures of NiO (950 °C, 2 h). The data for the glass-ceramic materials from Series 2 are marked with ■ and □. The lines are presented as guides for the eye.

compositions of Series 1.2.x, the density increased for composition 1.2.4, further additions of NiO reduced the density.

The mechanical strength of glass-ceramics produced from the glass compositions of Series 1.2.x (low [MgO]) improved for NiO admixtures greater than 4 wt%. The glasses of Series 1.1.x crystallized intensively, but deformed and/or self-fractured during the thermal treatment; for that reason they were not characterized mechanically.

The micrographs and XRD patterns of glass-ceramics of compositions 2.1.7 and 2.2.7 (analogues to 1.1.7 and 1.2.7) are presented in Figs. 7 and 8. The samples of the composition 2.2.7 (low [MgO]) treated at 800 °C (2 h) contained only bunsenite crystalline phase, whereas the treatment at 950 °C (2 h) promoted formation of the pyroxene-like crystals with the structure similar to diopside–hedenbergite solid solutions surrounding bunsenite nuclei.

The samples of the composition 2.1.7 treated at 800 °C also contained some of bunsenite crystals (Fig. 7) but in much lower content relative to composition 2.2.7; the pyroxene-like crystalline phase with the structure similar to $(\text{Mg}_{0.4}\text{Ni}_{0.6})(\text{Ca}_{0.9}\text{Ni}_{0.1})\text{Si}_2\text{O}_6$ was the main crystalline phase in these conditions. Increasing the temperature of treatment to 950 °C resulted in a more intensive formation of the pyroxene-like crystalline phase. In accordance with EDS spot analysis data, the NiO content in the crystals formed at 800 and 950 °C ranged at (wt%) 12.2–23.9 and 4.3–25.8, respectively.

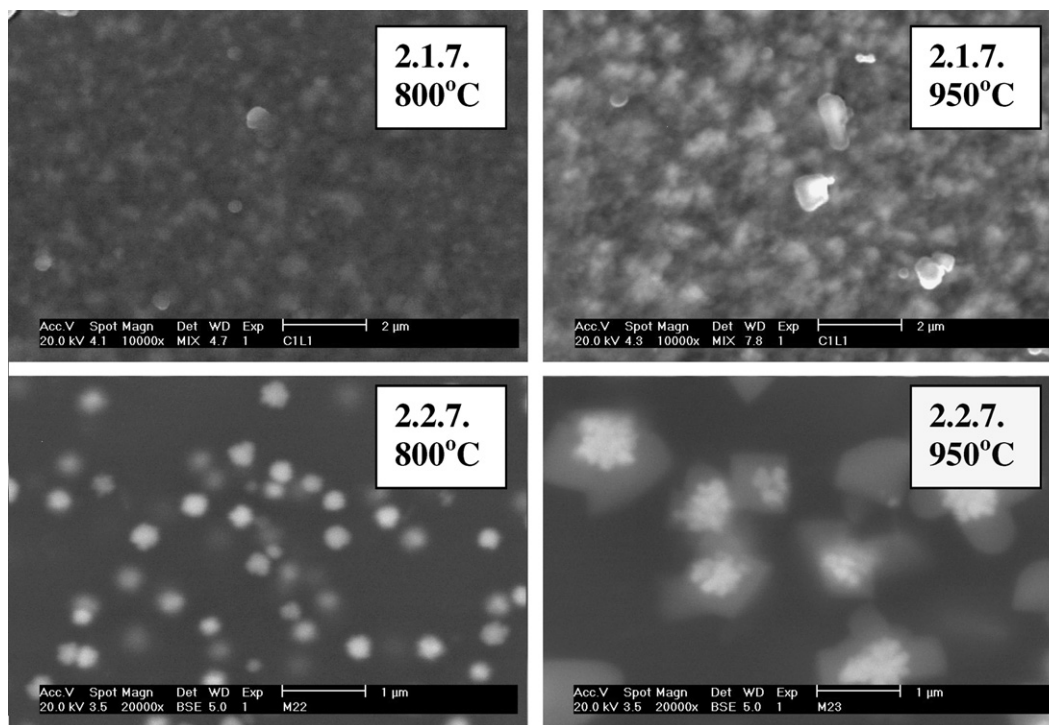


Fig. 7. Micrographs of glass-ceramics obtained with glass samples No 2.1.7 (magnification 10000×) and 2.2.7 (magnification 20000×) after the thermal treatment at 800 °C and 950 °C for 2 h.

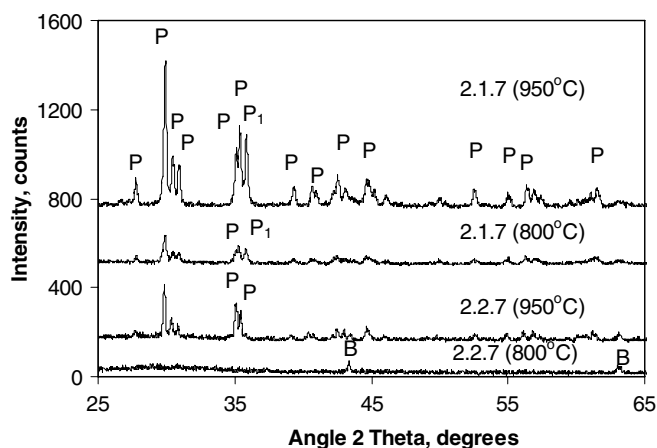


Fig. 8. X-ray diffractograms of the glass-ceramics produced from compositions of Series 2 at 800 and 950 °C (2 h). P – pyroxene-like crystalline phase similar to diopside–hedenbergite solid solutions, P₁ – characteristic reflection of the pyroxene-like crystalline phase similar to (MgO_{0.4}–NiO_{0.6})(CaO_{0.9}NiO_{0.1})Si₂O₆, B – bunsenite (NiO).

The glass-ceramics produced with industrial wastes (compositions 2.1.7 and 2.2.7) demonstrated densities, acid and alkali resistances similar to the model compositions 1.1.7 and 1.2.7, but much higher water resistance. The data on contents of Ni and Cr in the leachates obtained after the water resistance testing of glass-ceramic samples produced in Series 2 are reported in Table 4. Moreover, in contrast with the self-fracture noted for composition 1.1.7, the glass-ceramic of composition 2.1.7 preserved its structural integrity during the additional thermal treatment; it is also

Table 4

Contents of Ni and Cr in the leachates obtained after water resistance tests with the glass-ceramics 2.1.7 and 2.2.7

Glass-ceramic composition	Contents of element in the leachates (mg/l)	
	Ni	Cr
2.1.7	3.6 ± 0.3	0.007 ± 0.001
2.2.7	4.0 ± 0.2	0.010 ± 0.001

worth to note that compositions from series 2 showed high mechanical strength in the range of 160–180 MPa (Fig. 6).

4. Discussion

The introduction of NiO in the glass compositions traditional in manufacturing of pyroxene glass-ceramics, allows the production of glass-ceramic materials by regulated crystallization. The role of NiO in the structure depends on the MgO content.

In the compositions of Series 1.2.x and 2.2.7 (low [MgO]), NiO acts as a nucleating agent due to the early formation of bunsenite (Figs. 2, 7, 8), it supports the subsequent formation of a pyroxene-like crystalline phase with a structure similar to that of diopside–hedenbergite solid solutions and inhibits the crystallization of gehlenite and nepheline.

In the composition of Series 1.1.x and 2.1.7 (high [MgO]), the introduction of NiO also supported early crystallization of bunsenite; however, at much lower contents relative to the glass-ceramics formed in the compositions

with low [MgO]. The admixtures of 4–7 wt% of NiO promoted formation of the pyroxene-like crystals with different contents of NiO: a crystalline phase with a structure similar to $(\text{MgO}_{0.4}\text{NiO}_{0.6})(\text{CaO}_{0.9}\text{NiO}_{0.1})\text{Si}_2\text{O}_6$ appeared at low temperatures simultaneously with bunsenite, whereas the pyroxene-like crystals with lower NiO contents crystallized later (Figs. 1, 7, 8). This effect was especially clear in the compositions with 6–7 wt% of NiO. In the composition 1.1.10, the pyroxene-like crystalline phase appeared at higher temperature and its crystals indicated a smaller variation of [NiO] in comparison with the composition 1.1.7.

Thus, for the investigated glass compositions, those containing 4–7 wt% of NiO are more promising to produce glass-ceramics due to the early nucleation of bunsenite (compositions with low [MgO]) or bunsenite (NiO) altogether with pyroxene-like crystals similar to $(\text{MgO}_{0.4}\text{NiO}_{0.6})(\text{CaO}_{0.9}\text{NiO}_{0.1})\text{Si}_2\text{O}_6$. These compositions also displayed a good combination of exploitation properties, except mechanical strength for glass-ceramics with high MgO which self-fractured during the thermal treatment. However, it seems that the presence of Cr_2O_3 (compositions based on galvanic slurries) supported earlier crystallization of pyroxene-like crystals of low NiO contents, and the formation of a more homogeneous glass-ceramic material characterized with relatively high mechanical strength. The glass-ceramics based on the glass compositions with $[\text{NiO}] > 8$ wt% (composition 1.1.10) displayed reduced and not promising exploitation properties. The results have shown that glass-ceramic materials with highly attractive properties can be produced on the basis of combinations of Ni containing galvanic slurries with raw materials traditional for the glass industry, such as sand, limestone, dolomite and soda ash (compositions with relatively low [MgO]), or with other industrial wastes. For example, the combination of coal ash with dried Ni containing galvanic slurry (17.3 wt% in the batch composition 2.2.7) made it possible to obtain glass-ceramic materials which only required a small addition of soda ash; nonetheless, the most important aspect is the incorporation of high amount of wastes. The application properties of such glass-ceramic materials, especially water and mechanical resistance, were enhanced relative to materials obtained with a NiO source different from the galvanic slurries (Figs. 5 and 6). The results obtained for glass-ceramics from Series 2 indicated leachate concentrations of Ni and Cr lower than 11 and 0.6 mg/l, respectively, (regulatory limits). Thus, the produced materials can be regarded as non-hazardous.

5. Conclusions

1. Glasses with high NiO contents can be produced in the system of $\text{Na}_2\text{O}-\text{CaO}-\text{MgO}-\text{Fe}_2\text{O}_3-\text{Al}_2\text{O}_3-\text{SiO}_2$.

2. The crystallization behavior of these glasses depends on the NiO and MgO contents.
3. In the glass compositions with low MgO contents, increased [NiO] promoted low temperature formation of bunsenite crystals as nuclei for the subsequent crystallization of pyroxene-like crystalline phases with a structure similar to diopside–hedenbergite solid solutions instead of gehlenite and augite that formed in NiO free compositions.
4. In the compositions with high MgO content, NiO (in the range of 4–7 wt%) promotes early nucleation of bunsenite and pyroxene-like crystals with a structure similar to $(\text{MgO}_{0.4}\text{NiO}_{0.6})(\text{CaO}_{0.9}\text{NiO}_{0.1})\text{Si}_2\text{O}_6$. The following crystallization is accompanied with the formation of pyroxene-like crystals with a structure similar to diopside–hedenbergite solid solutions.
5. Glass-ceramic materials characterized with good application properties can be produced by combination of dried Ni-bearing galvanic slurry with other industrial wastes as raw materials.

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References

- [1] J.Ma. Rincon, M. Romero, A.R. Boccaccini, J. Mater. Sci. 34 (1999) 4413.
- [2] A. Gorokhovskiy, J.I. Escalante-Garcia, J. Mendez-Nonell, V. Gorokhovskiy, D. Mescheryakov, Am. Ceram. Soc. Bull. 80 (2001) 49.
- [3] C. Fredericci, E.D. Zanotto, E.C. Ziemath, J. Non-Cryst. Solids 273 (2000) 64.
- [4] A. Gorokhovskiy, D. Mescheryakov, V. Gorokhovskiy, J.I. Escalante-Garcia, M.I. Pech-Canul, Glaztech. Ber. Glass Sci. Technol. 73C1 (2000) 374.
- [5] D.U. Tulyaganov, M.J. Ribeiro, J.A. Labrincha, Ceram. Int. 28 (2002) 515.
- [6] L. Barbieri, C. Leonelli, T. Manfredini, G.C. Pellacani, C. Siligardi, E. Tondello, R. Bertinello, J. Mater. Sci. 29 (1994) 6273.
- [7] L.A. Zhunina, M.I. Kuzmenkov, V.N. Yaglov, Pyroxene glass-ceramics (Rus), Belorussian State University, Minsk, 1974.
- [8] F.J. Torres, J. Alarcon, J. Non-Cryst. Solids 320 (2003) 246.
- [9] A. Costantini, F. Branda, A. Buri, J. Mater. Sci. 30 (1995) 1561.
- [10] P. Alizadeh, V.K. Marghussian, J. Eur. Ceram. Soc. 20 (2000) 765.
- [11] L. Barbieri, A. Corradi, I. Lancellotti, J. Eur. Ceram. Soc. 22 (2002) 1759.
- [12] C. Leroy, M.C. Ferro, R.C.C. Monteiro, J. Eur. Ceram. Soc. 21 (2001) 195.
- [13] H. Endo, Y. Nagayoshi, K. Suzuki, Water Sci. Technol. 36 (1997) 235.
- [14] G.A. Khater, M.M. Morsi, K.J. Range, Glaztech. Ber. Glass Sci. Technol. 74 (2001) 192.
- [15] S.N. Salama, Mater. Chem. Phys. 28 (1991) 237.